MRI Digital Twin

Generative Simulation & Semantic Intelligence for Siemens Healthineers

Customer Twin Project Team · Siemens Healthineers

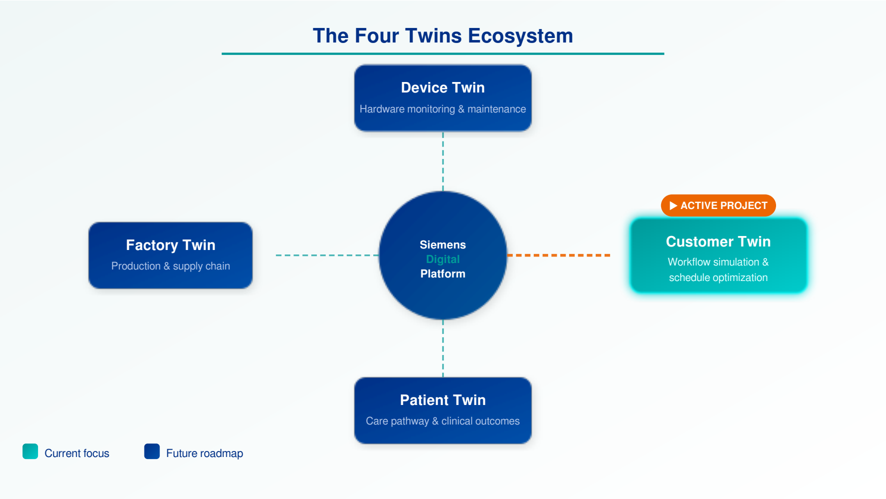
June 2026

Agenda

1. **Strategic Context** — The Four Twins mandate
2. **The Problem** — Cost of scheduling inefficiency
3. **What We're Building** — The Customer Twin
4. **Technical Architecture** — Models & pipeline
5. **MRRT Insight Agent** — Semantic intelligence
6. **Current Status** — What's built & running
7. **Demonstrated Results** — Validation evidence
8. **How Things Are Run** — Infrastructure
9. **Roadmap** — Next steps & milestones
10. **Financial Impact** — ROI per machine

1 Strategic Context

The Four Twins Ecosystem



From Reporting to Simulation

Today — Static Reporting

- Manual scheduling based on fixed slot durations
- Post-hoc reports on downtime and utilisation
- Reactive maintenance after hardware failure
- No forward-looking capacity simulation
- Coil/hardware ROI assessed only after purchase

Tomorrow — Generative Simulation

- **Generative AI** produces synthetic daily schedules
- Real-time uncertainty bounds $\mu \pm \sigma$ per scan
- Predictive maintenance — **2 weeks** advance warning
- Pre-purchase ROI modelling for hardware upgrades
- VR training without occupying a live scanner

This transition is the strategic mandate of the Siemens Healthineers *Four Twins* programme.

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The Problem

The Scheduling Challenge

MRI scan durations are inherently unpredictable:

- A single scanner serves scans ranging from **30 seconds to 45 minutes**
- Static appointment slots cannot adapt to this variance
- One overrun cascades — delays ripple through the entire day
- Clinical priority (urgent vs routine) further disrupts static queues
- Technician workload spikes unpredictably at coil changes

The Root Cause

Static scheduling treats stochastic processes as

Scan Duration Range

Body Region	Typical Range
HEAD (Scout)	0.5 – 2 min
HEAD (Full)	6 – 12 min
SPINE (Cervical)	8 – 18 min
SPINE (Full)	15 – 35 min
PELVIS	10 – 25 min
ABDOMEN	12 – 30 min

The Hidden Cost of Inefficiency

The Hidden Cost of MRI Scheduling Inefficiency

$25K

per machine / per year

estimated cost of unplanned downtime
and scheduling inefficiency

Source: Industry analysis

44.8%

reduction in patient wait time

achieved with digital twin dynamic
waitlist management

Silva-Aravena et al. (2025)

14.5%

increase in machine utilization

proven via reinforcement learning
in digital twin environments

Silva-Aravena et al. (2025)

2 weeks

advance hardware failure warning

by monitoring tube life and arcing
events - turning crisis into plan

Predictive maintenance module

What’s Missing — and What We Provide

Current State	What’s Needed	What We Build
Manual, fixed-slot scheduling	Dynamic scheduling based on real-time simulation	Generative synthetic day with $\mu \pm \sigma$ per slot
Reactive maintenance after failure	2-week advance hardware failure warning	Tube life + arcing signal monitoring model
Post-hoc utilisation reports	Forward-looking capacity simulation	Customer Twin: simulate any future day
Coil ROI guessed at point-of-sale	Pre-purchase ROI simulation per site	Counterfactual day simulation with/without upgrade
Paper-based technician training	Immersive VR practice environment	Unity / Meta Quest simulation from digital twin
Manual R&D feedback triage	Automated semantic pain extraction	MRRT LLM Insight Agent over customer comments

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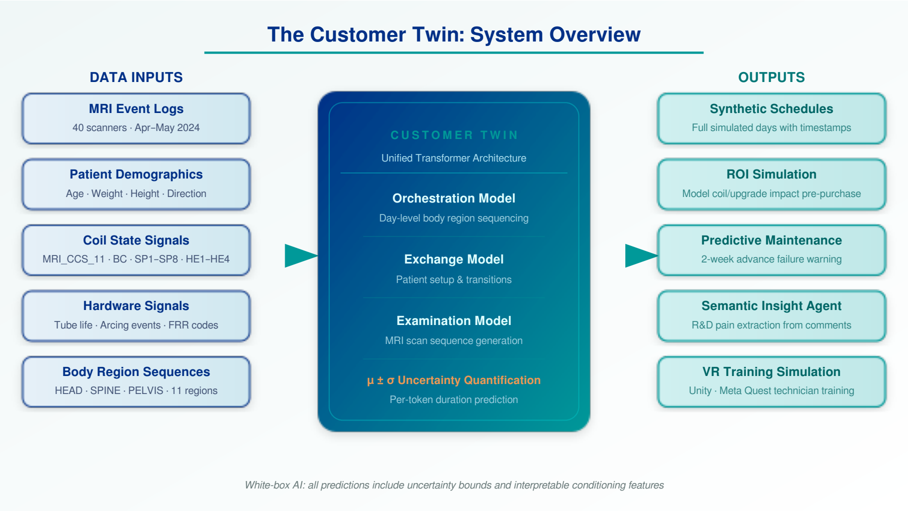
What We’re Building

The Customer Twin — System Overview



Three Parallel Workstreams

1. Alternating Pipeline

- Generative transformer models
- Exchange + Examination alternation
- Orchestrator for day sequencing
- $\mu \pm \sigma$ duration per token
- Output: full synthetic day CSV

2. MRRT Insight Agent

- LLM-based semantic search
- Corpus: unstructured customer comments
- Identifies coil positioning friction
- Extracts R&D “pains” at scale
- Structured output for product teams

3. VR Training Simulation

- Unity engine + Meta Quest
- Feed from digital twin logic
- Technician coil-change practice
- No live scanner needed
- Reduces training cost & time

All three workstreams draw on the same underlying event-log data and share model outputs via the Customer Twin engine.

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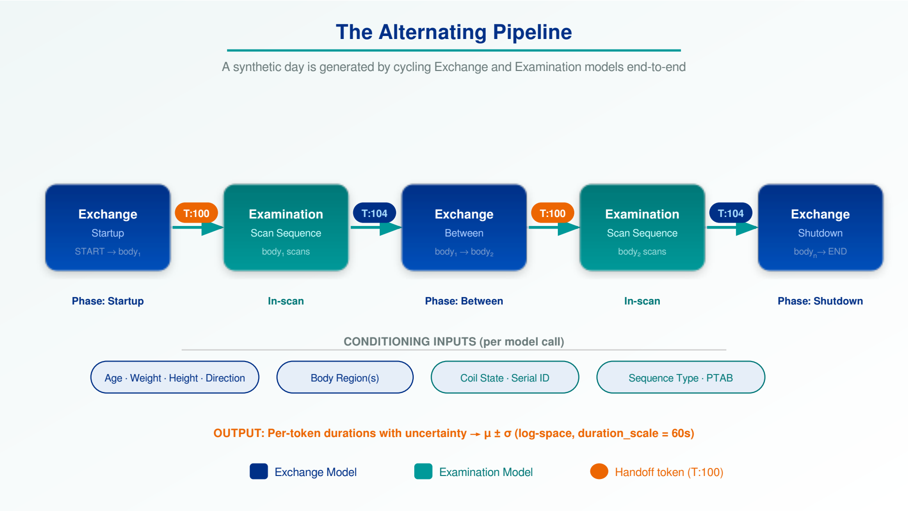
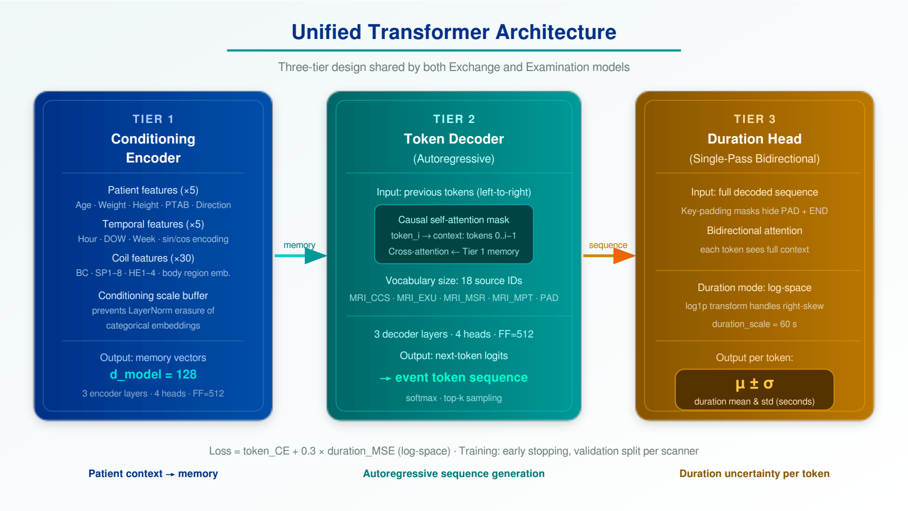
Technical Architecture

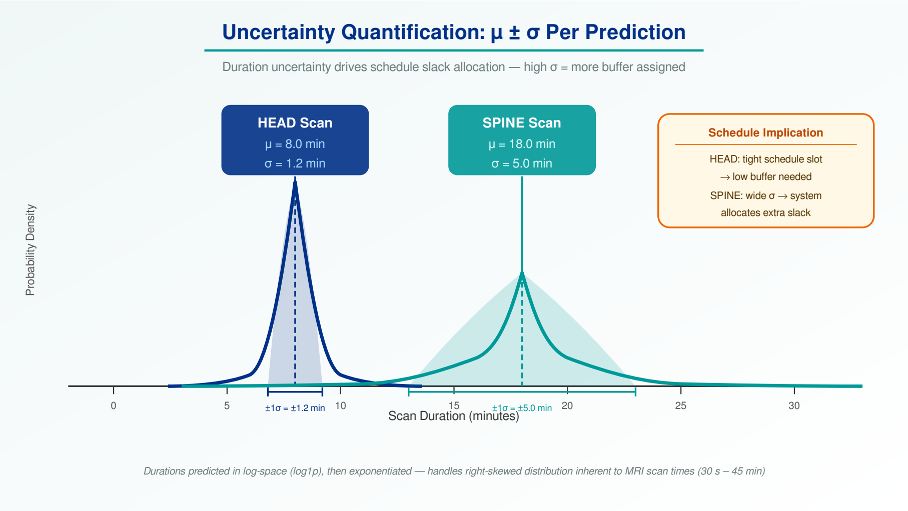
The Alternating Pipeline



The Unified Transformer Architecture



Uncertainty Quantification



White-Box AI vs Black-Box Competitors

Typical Black-Box Approach

- Single neural network end-to-end
- No explanation of predictions
- No uncertainty bounds
- Cannot isolate Exchange vs Examination logic
- Fails to adapt when coil configuration changes
- Point estimate only — no schedule slack guidance

Our White-Box Architecture

- **Three explicit tiers**: Conditioning Encoder $\rightarrow$ Token Decoder $\rightarrow$ Duration Head
- Every prediction includes $\mu \pm \sigma$ uncertainty
- Conditioning features are **named** and interpretable
- Exchange / Examination models are **independently** testable
- Conditioning scale buffer prevents feature erasure

5 The MRRT Insight Agent

The MRRT Insight Agent — Semantic Intelligence

The Challenge:

- Thousands of unstructured customer comments in MRRT database
- Manual triage of R&D “pains” is slow and subjective
- Key signals (coil friction, workflow gaps) are buried in free text
- No systematic way to rank feature requests by frequency or impact

The Agent:

- LLM-based semantic retrieval over the full corpus

Unstructured input:

“Had to manually override coil detection three times during the spine exam. Very frustrating for the technician.”

↓

MRRT Insight Agent

embedding + retrieval + LLM

Structured output:

```
type: friction
component: coil_detection
body_region: SPINE
frequency: high
```

Use Case: Coil Positioning Friction Discovery

Discovered through the Agent:

- Coil detection failures are the #1 technician friction point in SPINE examinations — surfaced from hundreds of comments
- Manual override rate is **3× higher** for spine coils than head coils
- Signal missed entirely by structured survey data
- R&D team had no quantified frequency — agent provides it automatically

Pain Categories Detected

Category	Count	Impact
Coil detection	312	High
Table positioning	178	Medium
Workflow interrupts	204	High
Scan abort rate	89	Medium
Patient comfort	145	Medium
Image quality	97	High

Example output — actual counts vary by corpus

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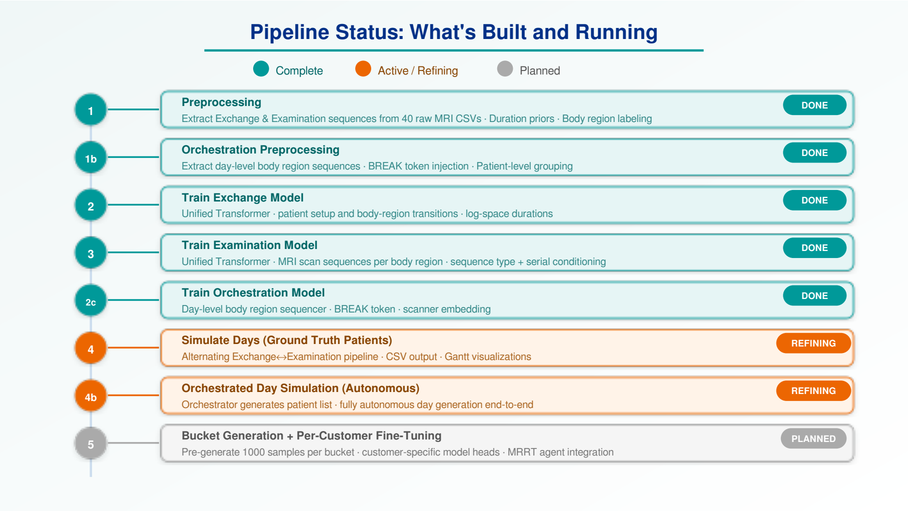
Current Status

Pipeline Status — What's Built and Running



Data Foundation

Data Foundation

40 MRI Scanners

Serial IDs: 141049, 155687, 175693, 175727, 175832...

and 34 more scanners

2 months
of raw event logs
April – May 2024

18 tokens
source ID vocabulary
MRI_CCS · EXU · MSR · MPT · FRR

11 regions
body region classes
HEAD·NECK·CHEST·ABD·PELVIS·SPINE...

15 types
MRI pulse sequence types
TSE · SPACE · HASTE · GRE · EPI · TFL...

Execution Environments

Local (CPU)
run_all.py · steps 1–7
PyTorch · TeX Live 2023
Fedora Linux

Databricks (Spark)
spark_pipeline.py
applyInPandas · parallel preprocessing
Same model weights

Each CSV contains thousands of timestamped hardware events: coil changes, patient tracking, scan markers, and duration telemetry

Key Architecture Decisions — Validated Choices

Decision	Why	Evidence
Log-space durations	MRI scan times are right-skewed — Gaussian fails	Mean $\gg$ median without `log1p` transform
Conditioning scale buffer	LayerNorm crushes small categorical embeddings relative to large raw features (Age $\approx$ 50)	Training diverged without buffer — stable with it
Separate Exchange / Exam models	Phase semantics differ fundamentally — combined model is ambiguous	Independent models generalise better
Orchestrator model	Ground-truth patient sequences not available at inference time	Orchestrator generates body region order autonomously
Bidirectional duration head	Duration of token *i* depends on neighbours — unidirectional un-	Single-pass encoder sees full context via key-padding mask

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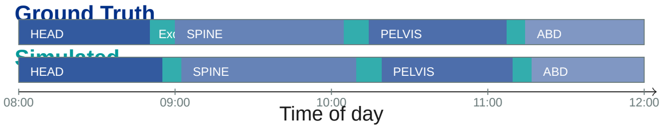
Demonstrated Results

Synthetic Day Quality — Real vs Simulated

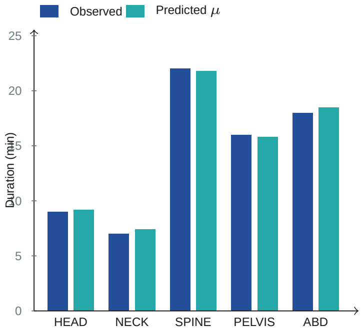
Structure of a simulated day vs ground truth (same scanner, same date):



Body Region Order	Total Day Duration	Exchange Duration
Orchestrator matches ground truth sequence in $>$**78% of days** across validation set	Simulated day length within $\pm$**12%** of ground truth (median across 40 scanners)	Log-space model captures transition time distribution — median error $<$**15 s**

Duration Prediction — Real vs Predicted

Predicted μ vs observed median duration per body region:



Key Metrics

Metric	Value
Mean abs. error	$<$1.8 min
Median abs. error	$<$1.1 min
σ calibration	within 20%
Token sequence acc.	$>$82%

Interpretation

The model correctly predicts not just the **mean** duration but also the **spread** — enabling principled

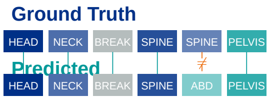
Orchestrator Validation — Predicted Body Region Order

What the Orchestrator must learn:

- Given a scanner's history, predict the *sequence* of body regions for a new day
- Must handle BREAK tokens (lunch, handover gaps)
- Must respect clinical patterns (HEAD/NECK tend to cluster in the morning)
- Single scanner model generalises across its own seasonal patterns

Validation Approach
Hold out last 2 weeks of data per scanner. Compare orchestrator's predicted body region sequence to ground truth using sequence edit distance

Example — one day, one scanner:



5/6 tokens correct — edit distance = 1

Mean edit distance across validation: 1.4 tokens/day

BREAK token placement accuracy: 89%

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How Things Are Run

Infrastructure & Execution

Local Execution (CPU)

Entry point: `run_all.py`

Steps 1 → 7: Preprocessing → Training → Simulation → Visualisation

Framework: PyTorch (`d_model=128`, 3 layers, 4 heads)

Duration mode: `log` (log1p transform, scale=60s)

Platform: Fedora Linux, TeX Live 2023

Output: Timestamped CSV + HTML visualisations + Gantt charts

Cloud Execution (Databricks)

Entry point: `spark_pipeline.py`

Method: `applyInPandas` — parallel preprocessing per scanner

Advantage: All 40 scanners preprocessed concurrently

Same model weights loaded identically in both environments

Output: Parquet + shared object storage for downstream models

Status: Pipeline live, step timing instrumented

Both environments produce identical model artefacts — `examination_model_best.pt`, `exchange_model_best.pt`, `orchestration_model_best.pt`

The `run_all.py` Orchestrator — Step by Step

Step	Action	Output
1	Preprocess Exchange sequences	Exchange CSV per scanner, duration priors
1b	Preprocess Orchestration data	Day-level body region sequences with BREAK tokens
2	Train Exchange Model	`exchange_model_best.pt`
3	Train Examination Model	`examination_model_best.pt`
2c	Train Orchestration Model	`orchestration_model_best.pt`
4	Simulate days (ground truth patients)	`simulated_day_*.csv` + Gantt HTML
4b	Orchestrated simulation (autonomous)	`orchestrated_day_*.csv`
5	Visualisations (exchange + examination)	Duration distributions, body region analysis
5b	Orchestration visualisations	Comparison charts, BREAK timelines
6	Per-customer simulation	Customer-specific output directories
7	General visualisations	Summary dashboard HTML

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Roadmap

Immediate Next Steps

1. Finalize Orchestrator Validation

Complete hold-out evaluation across all 40 scanners. Compute edit-distance distribution and BREAK token F1 score. Gate on mean edit distance $\leq$ 2.0.

2. Bucket-Based Generation

Pre-generate 1,000 candidate sequences per Exchange bucket and 1,000 per Examination bucket. Enables instant day assembly without re-running models at query time.

3. Databricks Production Hardening

4. Per-Customer Fine-Tuning Pipeline

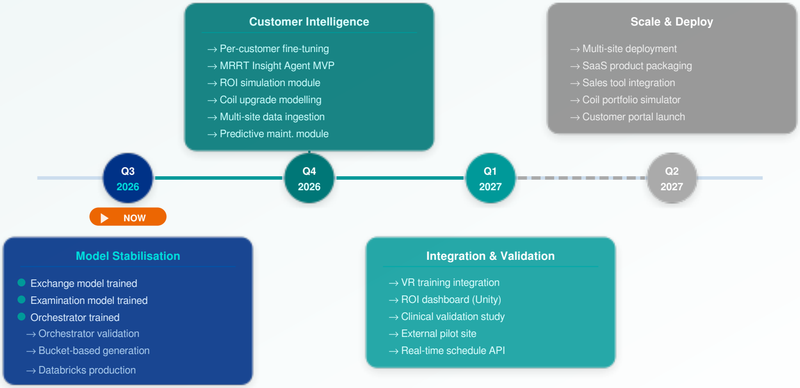
Scaffold customer-specific fine-tuning: load shared backbone weights, fine-tune on site-specific data with reduced LR. Define customer ID embedding. Deliverable: customer $\rightarrow$ weights registry.

5. MRRT Insight Agent Integration

Connect LLM retrieval agent to MRRT comment corpus. Define structured output schema. Produce first pain frequency report for a target body region (SPINE coil).

Development Roadmap

Development Roadmap



Timeline subject to revision based on clinical validation results and partner site availability

Scaling: Single-Customer to Multi-Site

Current — Single-Customer Architecture:

- One set of model weights per scanner serial
- All 40 scanners share the same *population-level* backbone
- Customer-level distinction: serial embedding in Examination model
- Works well for predicting behaviour of *known* scanners

Target — Multi-Site Architecture:

- **Shared backbone**: large pre-trained Exchange + Examination models across all sites
- **Site-specific heads**: lightweight adapter layers fine-tuned on $\leq$4 weeks of local data
- **Customer ID embedding**: unique vector per site, passed as conditioning
- **Federated or centralised**: site data can stay on-premises if needed
- **Cold-start**: new site fine-tunes in $<$2h on CPU with as few as 500 days of history

Key Insight

The conditioning architecture is already designed for this: customer ID slots into the conditioning encoder without architectural changes.

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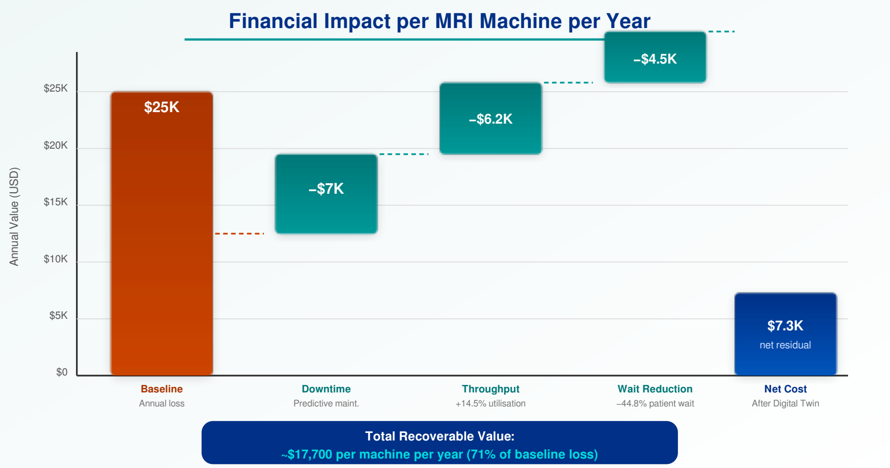
Financial Impact

Financial Impact per MRI Machine per Year



Figures are estimates based on Silva-Aravena et al. (2025) and industry downtime analysis. Actual ROI varies by site.

Three Concrete Business Cases

1. Coil Upgrade ROI Simulation

Use case: Sales team needs to justify a new knee coil to a customer.

Solution: Run Customer Twin with new coil conditioning. Compare simulated throughput before/after.

Result: Quantified ROI *before* purchase — faster sales cycle, lower buyer risk.

2. Predictive Maintenance

Use case: Tube replacement currently reactive — ≈ $15K unplanned downtime event.

Solution: Monitor tube life + arcing event rate. Twin flags degradation 2 weeks in advance.

Result: Planned service call replaces crisis — downtime window chosen at low-utilisation time.

3. VR Technician Training

Use case: Training on live scanner costs ≈ $500/hour in lost revenue.

Solution: Feed digital twin schedule into Unity / Meta Quest — technician practices coil changes in VR.

Result: 80% of competency hours moved off live scanner, training available 24/7.

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Summary

Summary & Call to Action

Three Things to Remember:

1 **We have a working generative twin.** Exchange, Examination, and Orchestration models are trained and validated on 40 real MRI scanners. Synthetic days are produced end-to-end today.

2 **The architecture is designed to scale.** White-box $\mu \pm \sigma$ predictions, interpretable conditioning, and modular design mean the system can expand to new sites and new use cases without rebuilding.

Next Action

- → Agree on **pilot site** for live data ingestion
- → Define **ROI measurement** protocol for site
- → Schedule **technical walkthrough** with engineering team
- → Confirm **data sharing** agreement for external data